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TEST ASSIST APPARATUS, TEST ASSIST METHOD, AND RECORDING MEDIUM

Abstract

This test assist apparatus includes: an acquiring section for acquiring utterance information which indicates the content of an utterance of a patient during a test performed on the patient, state information regarding the feelings of the patient during the test, and basic information regarding the test; an interruption predicting section for predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and an outputting section for outputting the probability of interruption.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-027605 filed on Feb. 27, 2024, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a test assist apparatus, a test assist method, and a recording medium.

BACKGROUND ART

[0003] Medical treatment assist techniques are known. Examples of medical treatment assist techniques include the technique disclosed in Patent Literature 1. Patent Literature 1 discloses a system for assisting medical treatments appropriate for various aspects of a patient. The system disclosed in Patent Literature 1 includes an acquiring section, an analyzing section, and a display control section. The acquiring section acquires patient information regarding a patient, medical interview information indicating the details of the patient's answers to the questions asked in a medical interview, and conversation information having recorded therein a conversation with the patient. The analyzing section performs analytical processing with use of the patient information, the medical interview information, and the conversation information, to generate patient characteristic information indicating the psychological aspect and societal aspect of the patient. The display control section displays, based on the patient characteristic information, the psychological aspect and the societal aspect of the patient such that the psychological aspect and the societal aspect are discriminable from each other.

CITATION LIST

Patent Literature

Patent Literature 1

[0004] Japanese Patent Application Publication Tokukai No. 2023-071244

SUMMARY OF INVENTION

Technical Problem

[0005] In some cases, a patient feels tense and uneasy in a test such as endoscopy. In such cases, the test can be interrupted because, for example, the patient feels a pain. With the technique disclosed in Patent Literature 1, there is a problem of being unable to predict an interruption of a test resulting from the feelings of a patient being tested, such as uneasiness.

[0006] The present disclosure has been made in view of the above problem, and an example object thereof is to provide a technique which makes it possible to predict an interruption of a test resulting from the feelings of a patient being tested, such as uneasiness.

Solution to Problem

[0007] A test assist apparatus in accordance with an example aspect of the present disclosure includes at least one processor, and the at least one processor carries out: an acquiring process of acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; an interruption predicting process of predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and an outputting process of outputting

the probability of interruption.

[0008] A test assist method in accordance with an example aspect of the present disclosure includes: at least one processor acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; the at least one processor predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and the at least one processor outputting the probability of interruption.

[0009] A recording medium in accordance with an example aspect of the present disclosure is a recording medium having recorded thereon a test assist program for causing a computer to function as a test assist apparatus, and the test assist program causes the computer to carry out: an acquiring process of acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; an interruption predicting process of predicting a probability of interruption of the test performed on the patient, from the utterance the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and an outputting process of outputting the probability of interruption.

Advantageous Effects of Invention

[0010] An example aspect of the present disclosure provides an example advantage of making it possible to provide a technique for predicting an interruption of a test resulting from the feelings of a patient being tested, such as uneasiness.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is a block diagram illustrating a configuration of a test assist apparatus in accordance with the present disclosure.

[0012] FIG. 2 is a flowchart illustrating a flow of a test assist method in accordance with the present disclosure.

[0013] FIG. 3 is a block diagram illustrating a configuration of an information processing apparatus in accordance with the present disclosure.

[0014] FIG. 4 is a diagram illustrating a functional configuration of a control section in accordance with the present disclosure.

[0015] FIG. 5 is a diagram illustrating a specific example of output information outputted by an output control section in accordance with the present disclosure.

[0016] FIG. 6 is a flowchart illustrating an example flow of a test assist method in accordance with the present disclosure.

[0017] FIG. 7 is a block diagram illustrating a configuration of a computer which functions as the test assist apparatus or the information processing apparatus in accordance with the present disclosure.

EXAMPLE EMBODIMENTS

[0018] The following description will discuss example embodiments of the present invention.

However, the present invention is not limited to the example embodiments described below, but can be altered by a skilled person in the art within the scope of the claims. For example, any embodiment derived by appropriately combining techniques (some or all of products or methods)

adopted in differing example embodiments described below can be within the scope of the present invention. Further, any embodiment derived by appropriately omitting one or more of the techniques adopted in differing example embodiments described below can be within the scope of the present invention. Furthermore, the advantage mentioned in each of the example embodiments described below is an example advantage expected in that example embodiment, and does not define the extension of the present invention. That is, any embodiment which does not provide the example advantages mentioned in the example embodiments described below can also be within the scope of the present invention.

First Example Embodiment

[0019] The following description will discuss a first example embodiment, which is an example embodiment of the present invention, in detail with reference to the drawings. The present example embodiment is basic to each of the example embodiments which will be described later. It should be noted that the applicability of the techniques adopted in the present example embodiment is not limited to the present example embodiment. That is, the techniques adopted in the present example embodiment can be adopted in another example embodiment included in the present disclosure, to the extent of constituting no specific technical obstacle. Further, the techniques illustrated in the drawings referred to for the description of the present example embodiment can be adopted in another example embodiment included in the present disclosure, to the extent of constituting no specific technical obstacle.

Configuration of Test Assist Apparatus

[0020] The configuration of a test assist apparatus **1** is described here with reference to FIG. **1**. FIG. **1** is a block diagram illustrating the configuration of the test assist apparatus **1**. The test assist apparatus **1** includes an acquiring section **11**, an interruption predicting section **12**, and an outputting section **13**, as illustrated in FIG. **1**.

[0021] The acquiring section **11** acquires utterance information indicating the content of an utterance of a patient during a test performed on the patient, state information regarding the feelings of the patient during the test, and basic information regarding the test. The interruption predicting section **12** predicts a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data. The outputting section **13** outputs the probability of interruption.

Example Advantage of Test Assist Apparatus

[0022] As above, the test assist apparatus **1** includes an acquiring section **11** for acquiring utterance information indicating the content of an utterance of a patient during a test performed on the patient, state information regarding the feelings of the patient during the test, and basic information regarding the test; an interruption predicting section **12** for predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and an outputting section **13** for outputting the probability of interruption. Thus, the test assist apparatus **1** provides an example advantage of making it possible to predict an interruption of a test resulting from the feelings of a patient being tested, such as uneasiness.

Flow of Test Assist Method

[0023] The flow of a test assist method **S1** is described here with reference to FIG. **2**. FIG. **2** is a flowchart illustrating the flow of the test assist method **S1**. The test assist method **S1** includes an acquiring process **S11**, an interruption predicting process **S12**, and an outputting process **S13**, as illustrated in FIG. **2**.

[0024] In the acquiring process **S11**, at least one processor acquires utterance information which

indicates the content of an utterance of a patient during a test performed on the patient, state information regarding the feelings of the patient during the test, and basic information regarding the test. In the interruption predicting process **S12**, the at least one processor predicts a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data. In the outputting process **S13**, the at least one processor outputs the probability of interruption.

Example Advantage of Test Assist Method

[0025] As above, the test assist method **S1** includes: an acquiring process **S11** of at least one processor acquiring utterance information which indicates the content of an utterance of a patient during a test performed on the patient, state information regarding the feelings of the patient during the test, and basic information regarding the test; an interruption predicting process **S12** of the at least one processor predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and an outputting process **S13** of the at least one processor outputting the probability of interruption. Thus, the test assist method **S1** provides an example advantage of making it possible to predict an interruption of a test resulting from the feelings of a patient being tested, such as uneasiness.

Second Example Embodiment

[0026] The following description will discuss a second example embodiment, which is an example embodiment of the present invention, in detail with reference to the drawings. A component having the same function as a component described in the above example embodiment is assigned the same reference sign, and the description thereof is omitted where appropriate. It should be noted that the applicability of the techniques adopted in the present example embodiment is not limited to the present example embodiment. That is, the techniques adopted in the present example embodiment can be adopted in another example embodiment included in the present disclosure, to the extent of constituting no specific technical obstacle. Further, the techniques illustrated in the drawings referred to for the description of the present example embodiment can be adopted in another example embodiment included in the present disclosure, to the extent of constituting no specific technical obstacle.

Configuration of Information Processing Apparatus

[0027] The configuration of an information processing apparatus **1A** is described here with reference to FIG. 3. FIG. 3 is a block diagram illustrating the configuration of the information processing apparatus **1A**. The information processing apparatus **1A** includes a control section **10A**, a storage section **20A**, a communicating section **30A**, an inputting section **40A**, and an outputting section **50A**.

Communicating Section

[0028] The communicating section **30A** communicates with an apparatus external to the information processing apparatus **1A** over a communication line. A specific configuration of the communication line does not limit the present example embodiment, but examples of the communication line include a wireless local area network (LAN), a wired LAN, a wide area network (WAN), a public network, a mobile data communication network, and a combination thereof. The communicating section **30A** transmits, to another apparatus, data supplied from the control section **10A**, and supplies the control section **10A** with data received from another apparatus.

Inputting Section

[0029] The inputting section **40A** is a component for accepting an input to the information

processing apparatus **1A**, and includes inputting equipment such as, for example, a keyboard, a mouse, a touch panel, a camera, or a microphone. Further, the inputting section **40A** may be a component for accepting data from inputting equipment via an interface such as, for example, a universal serial bus (USB).

Outputting Section

[0030] The outputting section **50A** is a component for producing an output from the information processing apparatus **1A**, and includes outputting equipment such as, for example, a display, a printer, a touch panel, or a speaker. The outputting section **50A** may be a component which, for example, includes an interface such as a USB and outputs data to outputting equipment via the interface.

Storage Section

[0031] In the storage section **20A**, various kinds of information to be referred to by the control section **10A** are stored. Examples of such information include medical information **201**, utterance information **202**, state information **203**, basic information **204**, probability of interruption **205**, and text **206**.

Medical Information

[0032] The medical information **201** is information regarding medical care for a patient. Examples of the medical information **201** include medical record information. As an example, the medical information **201** includes at least one selected from the group consisting of personal information regarding the patient, information on findings shown by a medical examination performed on the patient, and medical history information regarding the patient. The personal information regarding the patient is personal information regarding the patient, and includes, for example, information indicating the age, the gender, the amount of smoking, etc. of the patient. The information on findings shown by a medical examination performed on the patient is information indicating findings made by a medical service worker, such as a doctor, and includes, for example, information obtained by the medical service worker through a medical interview or an inspection. The information obtained through a medical interview or an inspection may be, for example, text “pressure on the abdomen causes a pain”. The medical history information regarding the patient is information regarding the history of disease of the patient, and includes, for example, information indicating the past medical history of the patient, the past medical history of the family, the amount of smoking of the patient, etc.

[0033] The medical information **201** may include a diagnosis target image of a patient. As an example, the diagnosis target image includes at least one of images which are an X-ray image, an endoscope image, a pathological image, an MRI image, and a CT image. In the storage section **20A**, respective pieces of medical information **201** regarding a plurality of patients are stored.

Utterance Information

[0034] The utterance information **202** is information indicating the content of an utterance of a patient during a test. For example, the utterance information **202** may be speech data obtained by picking up an utterance of a patient being tested. In this case, examples of the utterance of a patient includes, for example, a complaint and a question made by the patient. The utterance information **202** may be data in text format, or may be speech data. For example, the utterance information **202** is text representing a patient's utterance such as “Is it safe? I am scared. I feel a pain.”

State Information

[0035] The state information **203** is information regarding the feelings of a patient. As an example, the state information **203** includes information which indicates at least one selected from the group consisting of the facial expression, the manner of speaking, a vital sign, and a feelings analysis result of the patient in a test. As an example, the feelings analysis is carried out by a feelings analyzing section **14A**, which will be described later. In other words, the state information **203** may include information indicating the result of an analysis, carried out by the feelings analyzing section **14A**, of the feelings of the patient. As an example, the information indicating the result of

an analysis of the feelings indicates a state of “tenseness” or “uneasiness”.

Basic Information

[0036] The basic information **204** is information regarding a test. As an example, the basic information **204** includes information indicating the type of test (lower endoscopy, etc.) and the phase of test (insertion of an endoscope). Further, the basic information **204** may include information (e.g., “first-ever”, “second”, “third”, etc.) indicating whether a patient has an experience of undergoing the test.

Probability of Interruption

[0037] The probability of interruption **205** is a probability (e.g., **40%**) that a test will be interrupted. The probability of interruption **205** is a probability predicted by an interruption predicting section **12A**, which will be described later.

Text

[0038] The text **206** is text representing at least one selected from the group consisting of the basis for an interruption of a test performed on a patient and advice on dealing with the interruption. Examples of the text **206** include text “The patient has a strong feeling of uneasiness about undergoing the test for the first time. In some of past similar cases, the uneasy and tense feelings of a patient stiffened their abdomen, and the insertion of an endoscope therefore caused a pain. Additional explanation is highly likely to allay patient's uneasiness.”

Control Section

[0039] FIG. **4** is a diagram illustrating a functional configuration of the control section **10A**. The control section **10A** includes an acquiring section **11A**, an interruption predicting section **12A**, an output control section **13A**, and a feelings analyzing section **14A**.

Acquiring Section

[0040] The acquiring section **11A** acquires the utterance information **202** indicating the content of an utterance of a patient during a test performed on the patient and the basic information **204** regarding the test, and provides the interruption predicting section **12A** with the utterance information **202** and the basic information **204** acquired. As an example, the acquiring section **11A** may acquire the utterance information **202** and the basic information **204** by retrieving the utterance information **202** and the basic information **204** from a storage location (which may be storage in the information processing apparatus **1A**, or may be storage external to the information processing apparatus **1A**) designated by a user of the information processing apparatus **1A**. The acquiring section **11A** may acquire the utterance information **202** and the basic information **204** by receiving the utterance information **202** and the basic information **204** from another apparatus via the acquire the utterance information **202** and the basic information **204** inputted to the inputting section **40A**.

[0041] As an example, the acquiring section **11A** acquires speech data representing an utterance of a patient picked up during a test, and converts the speech data into the utterance information **202** in text format. In this case, the acquiring section **11A** may carry out speaker recognition based on the speech data and transcribe utterances speaker by speaker, to generate the utterance information **202** which indicates the content of an utterance of the patient.

[0042] Further, the acquiring section **11A** may acquire the medical information **201** in addition to the utterance information **202** and the basic information **204**. As an example, the acquiring section **11A** may acquire the medical information **201** by retrieving the medical information **201** from a storage location (which may be storage in the information processing apparatus **1A**, or may be storage external to the information processing apparatus **1A**) designated by a user of the information processing apparatus **1A**. The acquiring section **11A** may acquire the medical information **201** by receiving the medical information **201** from another apparatus via the acquire the medical information **201** inputted to the inputting section **40A**.

Feelings Analyzing Section

[0043] The feelings analyzing section **14A** analyzes the feelings of a patient in a test, and generates

the result of the analysis. As an example, the feelings analyzing section **14A** may analyze at least one selected from the group consisting of speech data regarding the patient picked up in a test and image data captured in the test, to analyze the feelings of the patient. The image data may be data representing a still image, or may be data representing a moving image. Examples of an approach for the feelings analysis may include an approach of using dictionaries prepared in advance and an approach of using a trained model generated by machine learning. In a case of using the trained model, the input to the trained model includes, for example, at least one selected from the group consisting of text, speech data, and image data. The output from the trained model includes information indicating the result of classification of the feelings.

[0044] The feelings analyzing section **14A** provides the interruption predicting section **12A** with the analysis result. The feelings analyzing section **14A** may output the analysis result by writing the analysis result in a storage location (which may be storage in the information processing apparatus **1A**, or may be storage external to the information processing apparatus **1A**) designated by a user of the information processing apparatus **1A**. Further, the feelings analyzing section **14A** may transmit the analysis result to another apparatus via the communicating section **30A**, or may output the analysis result to outputting equipment such as a display.

Interruption Predicting Section

[0045] The interruption predicting section **12A** acquires the state information **203**. As an example, the interruption predicting section **12A** may acquire the state information **203** by retrieving the state information **203** from a storage location (which may be storage in the information processing apparatus **1A**, or may be storage external to the information processing apparatus **1A**) designated by a user of the information processing apparatus **1A**. The interruption predicting section **12A** may acquire the state information **203** by receiving the state information **203** from another apparatus via the communicating section **30A**. The interruption predicting section **12A** may acquire the state information **203** inputted to the inputting section **40A**.

[0046] As an example, the state information **203** includes the analysis result provided by the feelings analyzing section **14A**. Further, the state information **203** may include information inputted by a medical service worker who performs a test on a patient, with use of the inputting section **40A**, or information transmitted to the information processing apparatus **1A** from a communication apparatus operated by the medical service worker. In this case, the medical service worker performing a test on a patient observes the facial expression, etc. of the patient in the test, and in a case where, for example, it is considered that the patient is feeling tense or uneasy and/or the patient does not understand the explanation, the medical service worker inputs information to that effect.

[0047] The state information **203** may include information inputted by the patient with use of the inputting section **40A**, or information transmitted to the information processing apparatus **1A** from a communication apparatus operated by the patient. In this case, if the patient cannot understand the explanation made by the medical service worker and/or the patient feels tense or uneasy about therapy, the patient inputs information to that effect.

[0048] The interruption predicting section **12A** predicts the probability of interruption **205** of a test performed on a patient, from the utterance information **202**, the state information **203**, and the basic information **204**, with use of a prediction model **M1** for predicting a probability of interruption of a test. The prediction model **M1** may be stored in the storage section **20A** of the information processing apparatus **1A**, or may be stored in an apparatus other than the information processing apparatus **1A**. The prediction model **M1** being stored in the storage section **20A** means that parameters defining the prediction model **M1** are stored in the storage section **20A**.

[0049] The prediction model **M1** is a model generated by machine learning in which samples of past test interruptions are used as training data. As an example, the training data includes a plurality of sets each of which is a set of (i) utterance information regarding a patient, the state information, and the basic information regarding a test in the test of the past and (ii) a label which indicates

whether the test was interrupted.

[0050] As an example, input information inputted by the interruption predicting section **12A** to the prediction model **M1** includes at least one selected from the group consisting of the utterance information **202** regarding a patient, the state information **203**, and the basic information **204** regarding a test in the test. Further, the input information inputted by the interruption predicting section **12A** to the prediction model **M1** may include the medical information **201**. Furthermore, the output from the prediction model **M1** includes the probability of interruption **205**.

[0051] The interruption predicting section **12A** generates, from the basic information **204**, the utterance information **202**, and the state information **203**, the text **206** representing at least one selected from the group consisting of the basis for an interruption of a test performed on a patient and advice on dealing with the interruption, with use of a language model **M2** generated by machine learning. For example, the language model **M2** is a large language model formed by an artificial neural network having a great number of parameters. The language model **M2** may be stored in the storage section **20A** of the information processing apparatus **1A**, or may be stored in an apparatus other than the information processing apparatus **1A**. The language model **M2** being stored in the storage section **20A** means that parameters defining the language model **M2** are stored in the storage section **20A**.

[0052] Examples of the language model **M2** includes, but is not limited to, generative AI such as Chat Generative Pre-Trained Transformer (ChatGPT) or Generative Pre-trained Transformer 4 (GPT-4), and such generative AI having been fine-tuned with data related to medical care.

[0053] Input information inputted by the interruption predicting section **12A** to the language model **M2** includes at least one selected from the group consisting of the utterance information **202**, the state information **203**, and the basic information **204**. Further, the input information may include the medical information **201**. Furthermore, the input information may include information indicating instructions that the text **206** is to be generated. For example, this information may be text “please tell me the basis for an interruption of the test and advice on dealing with the interruption.” Output information outputted from the language model **M2** includes the text **206**.

[0054] In a case where the language model **M2** is stored in an apparatus other than the information processing apparatus **1A**, the interruption predicting section **12A** inputs the input information to the language model **M2** by, for example, transmitting, via the communicating section **30A**, the input information to the apparatus in which the language model **M2** is stored. In this case, the interruption predicting section **12A** receives the information outputted by the language model **M2** from that apparatus via the communicating section **30A**.

[0055] The interruption predicting section **12A** may refer to the state information **203** to determine whether the patient is feeling uneasy. In this case, the interruption predicting section **12A** may refer to the state information **203** to determine whether the patient is feeling uneasy, and in a case of a determination that the patient is feeling uneasy, the interruption predicting section **12A** may predict the probability of interruption and generate the text **206**.

Output Control Section

[0056] The output control section **13A** outputs output information **207**, which includes the probability of interruption **205** predicted by the interruption predicting section **12A** and the text **206**. As an example, the output control section **13A** may output the probability of interruption **205** and the text **206** by writing the probability of interruption **205** and the text **206** in a storage location (which may be storage in the information processing apparatus **1A**, or may be storage external to the information processing apparatus **1A**) designated by a user of the information processing apparatus **1A**. Further, the output control section **13A** may transmit the probability of interruption **205** and the text **206** via the communicating section **30A**, or may output the probability of interruption **205** and the text **206** to outputting equipment such as a display.

[0057] FIG. 5 is a diagram illustrating specific examples of the probability of interruption **205** and the text **206** which are outputted by the output control section **13A**. As an example, the output

control section **13A** may display the output information **207**, which includes the probability of interruption **205** and the text **206**, on displaying equipment, as illustrated in FIG. 5. In this case, a screen displayed on the displaying equipment includes the probability of interruption **205** and the text **206**. The screen further includes the basic information **204**, the state information **203**, and the utterance information **202**, which are inputs to the information processing apparatus **1A**. As illustrated in FIG. 5, the output control section **13A** may output at least one selected from the group consisting of the basic information **204**, the state information **203**, and the utterance information **202**, together with the probability of interruption **205** and the text **206**.

[0058] In the example of FIG. 5, the output information **207** includes the basic information **204** in which the “details of test” is “Lower endoscopy for the first time. Insertion of endoscope.” Further, the output information **207** includes the state information **203** in which the “facial expression and attitude” is “tenseness/uneasiness”. Further, the output information **207** includes the utterance information **202** in which the “complaint/question” is “Is it safe? I am scared. I feel a pain.”

[0059] In the example of FIG. 5, the output information **207** includes the probability of interruption **205** in which the “interruption/continuation determination” is “Probability that test will be interrupted: 40%”. Further, the output information **207** includes the text **206** in which the “basis” is “The patient has a strong feeling of uneasiness about undergoing the test for the first time. In some of past similar cases, the uneasy and tense feelings of a patient stiffened their abdomen, and the insertion of an endoscope therefore caused a pain. Additional explanation is highly likely to allay patient's uneasiness.”

[0060] For example, at least one selected from the group consisting of the probability of interruption **205** and the text **206** is used in decision-making by a medical service worker who performs a test on a patient. As an example, the medical service worker checks the probability of interruption **205** outputted, to decide on an action such as temporarily interrupting the test to make an additional explanation with the text **206** used as a guide, in order to allay the uneasiness of the patient.

Flow of Test Assist Method

[0061] FIG. 6 is a flowchart illustrating an example flow of a test assist method **S1A** carried out by the information processing apparatus **1A**. The steps included in the flowchart of FIG. 6 may be carried out in parallel with each other or in a different order. In step **S101**, the feelings analyzing section **14A** analyzes the facial expression, the content of a speech, and the like of a patient in a test, to generate an analysis result, which is the state information **203**.

[0062] In step **S102**, the feelings analyzing section **14A** outputs the state information **203**, which is the analysis result, to outputting equipment. As an example, the feelings analyzing section **14A** may output the state information **203** to a display, and display the state information **203** on the display.

[0063] A medical service worker who performs a test on the patient checks the state information **203** displayed on the display, to determine whether to predict the probability of interruption **205**. In a case of predicting the probability of interruption **205**, the medical service worker operates an operation element or the like which is connected to the inputting section **40A**, to input, to the information processing apparatus **1A**, an instruction to predict the probability of interruption **205**. For example, in a case where the state information **203** outputted includes at least one selected from the group consisting of information indicating “tenseness” and information indicating “uneasiness”, the medical service worker determines that the probability of interruption **205** should be predicted. In a case of a determination that the probability of interruption **205** does not need to be predicted, the medical service worker does not input the prediction instruction. For example, in a case where the state information **203** outputted does not include information indicating “tenseness” or information indicating “uneasiness”, the medical service worker determines that the probability of interruption **205** should not be predicted.

[0064] In step **S103**, the interruption predicting section **12A** determines whether to predict the probability of interruption **205**. As an example, the interruption predicting section **12A** makes this

determination according to whether the prediction instruction is inputted by a user such as the medical service worker or the like. In a case of a determination that the probability of interruption **205** should be predicted, the interruption predicting section **12A** moves to the process of step **S104**. In a case of a determination that the probability of interruption **205** should not be predicted, the interruption predicting section **12A** skips the processes of step **S104** to step **S109** and returns to the process of step **S101**.

[0065] A determination approach carried out by the interruption predicting section **12A** in step **S103** is not limited to the above-described approach, but the interruption predicting section **12A** may use another approach to determine whether to predict the probability of interruption **205**. As an example, the interruption predicting section **12A** may refer to the state information **203** generated by the feelings analyzing section **14A**, to determine whether to predict the probability of interruption **205**. In this case, as an example, in a case where the state information **203** includes at least one selected from the group consisting of information indicating “tenseness” and information indicating “uneasiness”, the interruption predicting section **12A** may determine that the probability of interruption **205** should be predicted. Further, as an example, in a case where the state information **203** does not include information indicating “tenseness” or information indicating “uneasiness”, the interruption predicting section **12A** may determine that the probability of interruption **205** should not be predicted.

[0066] In step **S104**, the acquiring section **11A** acquires the basic information **204** from the storage section **20A**. In step **S105**, the acquiring section **11A** acquires speech data picked up in a test. The speech data is data representing the content of an utterance of a patient during a test. In step **S106**, the acquiring section **11A** performs speaker recognition on the speech data and transcribes utterances speaker by speaker, to generate the utterance information **202** which indicates the content of an utterance of the patient. The acquiring section **11A** provides the interruption predicting section **12A** with the basic information **204** and the utterance information **202** generated.

[0067] In step **S107**, the interruption predicting section **12A** predicts the probability of interruption **205**, by inputting the basic information **204**, the utterance information **202**, and the state information **203** to the prediction model **M1**. Further, in step **S108**, the interruption predicting section **12A** generates the text **206**, by inputting the basic information **204**, the utterance information **202**, and the state information **203** to the language model **M2**. In step **S109**, the output control section **13A** outputs the probability of interruption **205** and the text **206**. Upon the completion of the process of step **S109**, the control section **10A** returns to the process of step **S101**.

Example Advantage of Information Processing Apparatus

[0068] As above, the information processing apparatus **1A** includes an interruption predicting section **12A** for generating, from the basic information **204**, the utterance information **202**, and the state information **203**, text **206** which indicates at least one selected from the group consisting of the basis for an interruption of a test performed on a patient and advice on dealing with the interruption, with use of a language model **M2** generated by machine learning, and an output control section **13A** outputs the text **206** together with the probability of interruption **205**. By, for example, a medical service worker referring to the text **206** outputted and interrupting a test to provide a patient with an additional explanation, it is possible to alleviate uneasiness or tenseness of the patient being tested. This makes it possible to prevent occurrence of a case where, for example, in endoscopy, the uneasy and tense feelings of a patient stiffen their abdomen, and the insertion of an endoscope therefore causes a pain. As above, the information processing apparatus **1A** provides an example advantage of making it possible to more smoothly proceed with a test than in a case of proceeding with the test without making an additional explanation.

[0069] The information processing apparatus **1A** includes a feelings analyzing section **14A** for analyzing the feelings of a patient during a test, and the interruption predicting section **12A** acquires the state information **203** which includes the analysis result provided by the feelings analyzing section **14A**. Thus, with the information processing apparatus **1A**, it is possible to make

an interruption prediction which incorporates the result of analyzing the feelings of a patient being tested, and by a medical service worker or the like checking the probability of interruption **205** outputted, it is possible to make a response such as temporarily interrupting the test to provide the patient with an additional explanation. This provides an example advantage of, for example, making it possible to more smoothly proceed with a test.

[0070] In the information processing apparatus **1A**, the acquiring section **11A** acquires speech data which represents a speech picked up during a test performed on a patient, and converts the speech data into the utterance information **202** in text format. Thus, the information processing apparatus **1A** provides an example advantage of making it possible to make an interruption prediction which incorporates the content of an utterance of a patient being tested.

[0071] In the information processing apparatus **1A**, the interruption predicting section **12A** refers to the state information **203** to determine whether a patient is feeling uneasy, and predicts the probability of interruption **205** in a case of a determination that the patient is feeling uneasy. Thus, with the information processing apparatus **1A**, it is possible to prevent interruption prediction to be made unnecessarily.

[0072] In the information processing apparatus **1A**, the state information **203** includes at least one selected from the group consisting of the facial expression, the manner of speaking, a vital sign, and a feelings analysis result of a patient in a test. Thus, with the information processing apparatus **1A**, it is possible to make an interruption prediction which incorporates information indicating at least one selected from the group consisting of the facial expression, the manner of speaking, a vital sign, and a feelings analysis result of a patient being tested.

[0073] In the information processing apparatus **1A**, the probability of interruption **205** is used in decision-making by a medical service worker who performs a test on a patient. Thus, the information processing apparatus **1A** provides an example advantage of making it possible for a medical service worker who performs a test to more properly carry out decision-making.

Software Implementation Example

[0074] Some or all of the functions of the test assist apparatus **1** and the information processing apparatus **1A** (hereinafter, also referred to as “each apparatus above”) may be implemented by hardware such as an integrated circuit (IC chip), or may be implemented by software.

[0075] In the latter case, the each apparatus above is provided by, for example, a computer that executes instructions of a program that is software implementing the functions. An example (hereinafter, computer C) of such a computer is illustrated in FIG. 7. FIG. 7 is a block diagram illustrating a hardware configuration of the computer C which functions as each apparatus above.

[0076] The computer C includes at least one processor C1 and at least one memory C2. The memory C2 has recorded thereon a program P for causing the computer C to operate as each apparatus above. The processor C1 of the computer C retrieves the program P from the memory C2 and executes the program P, so that the functions of each apparatus above are implemented.

[0077] Examples of the processor C1 can include a central processing unit (CPU), a graphic processing unit (GPU), a digital signal processor (DSP), a micro processing unit (MPU), a floating point number processing unit (FPU), a physics processing unit (PPU), a tensor processing unit (TPU), a quantum processor, a microcontroller, and a combination thereof. Examples of the memory C2 can include a flash memory, a hard disk drive (HDD), a solid state drive (SSD), and a combination thereof.

[0078] The computer C may further include a random access memory (RAM) into which the program P is loaded at the time of execution and in which various kinds of data are temporarily stored. The computer C may further include a communication interface via which data is transmitted to and received from another apparatus. The computer C may further include an input-output interface via which inputting-outputting equipment such as a keyboard, a mouse, a display, or a printer is connected.

[0079] The program P can be recorded on a non-transitory tangible recording medium M capable of

being read by the computer C. Examples of such a recording medium M can tape, a disk, a card, a semiconductor memory, and a programmable logic circuit. The computer C can obtain the program P via such a recording medium M. The program P can be transmitted via a transmission medium. Examples of such a transmission medium can include a communication network and a broadcast wave. The computer C can obtain the program P also via such a transmission medium.

[0080] The above-described functions of each apparatus above may be implemented by a single processor provided in a single computer, may be implemented by the cooperation among a plurality of processors provided in a single computer, or may be implemented by the cooperation among a plurality of processors provided in a plurality of respective computers. Further, the program for causing each apparatus above to implement the above-described functions may be stored in a single memory provided in a single computer, may be stored in a distributed manner in a plurality of memories provided in a single computer, or may be stored in a distributed manner in a plurality of memories provided in a plurality of respective computers.

Additional Remark 1

[0081] The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes, and the present invention can be altered in various ways by a skilled person in the art within the scope of the claims.

Additional Remark A

[0082] The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes, and the present invention can be altered in various ways by a skilled person in the art within the scope of the claims.

Supplementary Note A1

[0083] A test assist apparatus, including: [0084] an acquiring means for acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; [0085] an interruption predicting means for predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and [0086] an outputting means for outputting the probability of interruption.

Supplementary Note A2

[0087] The test assist apparatus described in supplementary note A1, further including [0088] a text generating means for generating, from the basic information, the utterance information, and the state information, text which represents at least one selected from the group consisting of basis for an interruption of a test performed on the patient and advice on dealing with the interruption, with use of a language model generated by machine learning, [0089] the outputting means being configured to output the text in addition to the probability of interruption.

Supplementary Note A3

[0090] The test assist apparatus described in supplementary note A1 or A2, further including [0091] a feelings analyzing means for analyzing feelings of the patient in the test, [0092] the acquiring means being configured to acquire the state information which includes an analysis result provided by the feelings analyzing means.

Supplementary Note A4

[0093] The test assist apparatus described in any one of supplementary notes A1 to A3, in which [0094] the acquiring means is configured to acquire speech data representing a speech picked up during the test performed on the patient, and convert the speech data into the utterance information in text format.

Supplementary Note A5

[0095] The test assist apparatus described in any one of supplementary notes A1 to A4, in which

the interruption predicting means is configured to refer to the state information to determine whether the patient is feeling uneasy, and predict the probability of interruption in a case of a determination that the patient is feeling uneasy.

Supplementary Note A6

[0096] The test assist apparatus described in any one of supplementary notes A1 to A5, in which [0097] the state information includes information which indicates at least one selected from the group consisting of a facial expression, a manner of speaking, a vital sign, and a feelings analysis result of the patient in the test.

Supplementary Note A7

[0098] The test assist apparatus described in any one of supplementary notes A1 to A6, in which [0099] the probability of interruption is used in decision-making by a medical service worker who performs a test on the patient.

Additional Remark B

[0100] The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes, and the present invention can be altered in various ways by a skilled person in the art within the scope of the claims.

Supplementary Note B1

[0101] A test assist method, including: [0102] at least one processor acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; [0103] the at least one processor predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and [0104] the at least one processor outputting the probability of interruption.

Supplementary Note B2

[0105] The test assist method described in supplementary note B1, further including [0106] the at least one processor generating, from the basic information, the utterance information, and the state information, text which represents at least one selected from the group consisting of basis for an interruption of a test performed on the patient and advice on dealing with the interruption, with use of a language model generated by machine learning, [0107] in the outputting, the at least one processor outputting the text in addition to the probability of interruption.

Supplementary Note B3

[0108] The test assist method described in supplementary note B1 or B2, further including [0109] the at least one processor analyzing feelings of the patient in the test, [0110] in the acquiring, the at least one processor acquiring the state information which includes an analysis result provided by the analyzing.

Supplementary Note B4

[0111] The test assist method described in any one of supplementary notes B1 to B3, in which [0112] in the acquiring, the at least one processor acquires speech data representing a speech picked up during the test performed on the patient, and converts the speech data into the utterance information in text format.

Supplementary Note B5

[0113] The test assist method described in any one of supplementary notes B1 to B4, in which [0114] in the predicting, the at least one processor refers to the state information to determine whether the patient is feeling uneasy, and predicts the probability of interruption in a case of a determination that the patient is feeling uneasy.

Supplementary Note B6

[0115] The test assist method described in any one of supplementary notes B1 to B5, in which

[0116] the state information includes information which indicates at least one selected from the group consisting of a facial expression, a manner of speaking, a vital sign, and a feelings analysis result of the patient in the test.

Supplementary Note B7

[0117] The test assist method described in any one of supplementary notes B1 to B6, in which [0118] the probability of interruption is used in decision-making by a medical service worker who performs a test on the patient.

Additional Remark C

[0119] The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes, and the present invention can be altered in various ways by a skilled person in the art within the scope of the claims.

Supplementary Note C1

[0120] A test assist program for causing a computer to function as a test assist apparatus, [0121] the test assist program causing the computer to function as: [0122] an acquiring means for acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; [0123] an interruption predicting means for predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and [0124] an outputting means for outputting the probability of interruption.

Supplementary Note C2

[0125] The test assist program described in supplementary note C1, further causing the computer to function as [0126] a text generating means for generating, from the basic information, the utterance information, and the state information, text which represents at least one selected from the group consisting of basis for an interruption of a test performed on the patient and advice on dealing with the interruption, with use of a language model generated by machine learning, [0127] the outputting means being configured to output the text in addition to the probability of interruption.

Supplementary Note C3

[0128] The test assist program described in supplementary note C1 or C2, further causing the computer function as [0129] a feelings analyzing means for analyzing feelings of the patient in the test, [0130] the acquiring means being configured to acquire the state information which includes an analysis result provided by the feelings analyzing means.

Supplementary Note C4

[0131] The test assist program described in any one of supplementary notes C1 to C3, in which [0132] the acquiring means is configured to acquire speech data representing a speech picked up during the test performed on the patient, and convert the speech data into the utterance information in text format.

Supplementary Note C5

[0133] The test assist program described in any one of supplementary notes C1 to C4, in which [0134] the interruption predicting means is configured to refer to the state information to determine whether the patient is feeling uneasy, and predict the probability of interruption in a case of a determination that the patient is feeling uneasy.

Supplementary Note C6

[0135] The test assist program described in any one of supplementary notes C1 to C5, in which [0136] the state information includes information which indicates at least one selected from the group consisting of a facial expression, a manner of speaking, a vital sign, and a feelings analysis result of the patient in the test.

Supplementary Note C7

[0137] The test assist program described in any one of supplementary notes C1 to C6, in which [0138] the probability of interruption is used in decision-making by a medical service worker who performs a test on the patient.

Additional Remark D

[0139] The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes, and the present invention can be altered in various ways by a skilled person in the art within the scope of the claims.

Supplementary Note D1

[0140] A test assist apparatus, including at least one processor, the at least one processor carrying out: [0141] an acquiring process of acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; [0142] an interruption predicting process of predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and [0143] an outputting process of outputting the probability of interruption.

[0144] The test assist apparatus may further include a memory. The memory may have stored therein a program for causing the at least one processor to carry out each of the processes.

Supplementary Note D2

[0145] The test assist apparatus described in supplementary note D1, in which the at least one processor further carries out [0146] a text generating process of generating, from the basic information, the utterance information, and the state information, text which represents at least one selected from the group consisting of basis for an interruption of a test performed on the patient and advice on dealing with the interruption, with use of a language model generated by machine learning, and [0147] in the outputting process, the at least one processor outputs the text in addition to the probability of interruption.

Supplementary Note D3

[0148] The test assist apparatus described in supplementary note D1 or D2, in which the at least one processor further carries out [0149] a feelings analyzing process of analyzing feelings of the patient in the test, and [0150] in the acquiring process, the at least one processor acquires the state information which includes an analysis result provided by the feelings analyzing process.

Supplementary Note D4

[0151] The test assist apparatus described in any one of supplementary notes D1 to D3, in which [0152] in the acquiring process, the at least one processor acquires speech data representing a speech picked up during the test performed on the patient, and converts the speech data into the utterance information in text format.

Supplementary Note D5

[0153] The test assist apparatus described in any one of supplementary notes D1 to D4, in which [0154] in the interruption predicting process, the at least one processor refers to the state information to determine whether the patient is feeling uneasy, and predicts the probability of interruption in a case of a determination that the patient is feeling uneasy.

Supplementary Note D6

[0155] The test assist apparatus described in any one of supplementary notes D1 to D5, in which [0156] the state information includes information which indicates at least one selected from the group consisting of a facial expression, a manner of speaking, a vital sign, and a feelings analysis result of the patient in the test.

Supplementary Note D7

[0157] The test assist apparatus described in any one of supplementary notes D1 to D6, in which [0158] the probability of interruption is used in decision-making by a medical service worker who

performs a test on the patient.

Additional Remark E

[0159] The whole or part 41 the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes, and the present invention can be altered in various ways by a skilled person in the art within the scope of the claims.

Supplementary Note E1

[0160] A non-transitory recording medium having recorded thereon a test assist program for causing a computer to function as a test assist apparatus, [0161] the test assist program causing the computer to carry out: [0162] an acquiring process of acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; [0163] an interruption predicting process of predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and [0164] an outputting process of outputting the probability of interruption.

Reference Signs List

[0165] **1**: Test assist apparatus [0166] **1A**: Information processing apparatus [0167] **11**, **11A**: Acquiring section [0168] **12**, **12A**: Interruption predicting section [0169] **13**, **50A**: Outputting section [0170] **13A**: Output control section [0171] **14A**: Feelings analyzing section

Claims

1. A test assist apparatus, comprising at least one processor, the at least one processor carrying out: an acquiring process of acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; an interruption predicting process of predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and an outputting process of outputting the probability of interruption.
2. The test assist apparatus according to claim 1, the at least one processor further carries out a text generating process of generating, from the basic information, the utterance information, and the state information, text which represents at least one selected from the group consisting of basis for an interruption of a test performed on the patient and advice on dealing with the interruption, with use of a language model generated by machine learning, and in the outputting process, the at least one processor outputs the text in addition to the probability of interruption.
3. The test assist apparatus according to claim 1, wherein the at least one processor further carries out a feelings analyzing process of analyzing feelings of the patient in the test, and in the acquiring process, the at least one processor acquires the state information which includes an analysis result provided by the feelings analyzing process.
4. The test assist apparatus according to claim 1, in the acquiring process, the at least one processor acquires speech data representing a speech picked up during the test performed on the patient, and converts the speech data into the utterance information in text format.
5. The test assist apparatus according to claim 1, wherein in the interruption predicting process, the at least one processor refers to the state information to determine whether the patient is feeling uneasy, and predicts the probability of interruption in a case of a determination that the patient is feeling uneasy.

6. The test assist apparatus according to claim 1, wherein the state information includes information which indicates at least one selected from the group consisting of a facial expression, a manner of speaking, a vital sign, and a feelings analysis result of the patient in the test.

7. The test assist apparatus according to claim 1, the probability of interruption is used in decision-making by a medical service worker who performs a test on the patient.

8. A test assist method, comprising: at least one processor acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; the at least one processor predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and the at least one processor outputting the probability of interruption.

9. A computer-readable non-transitory recording medium having recorded thereon a test assist program for causing a computer to function as a test assist apparatus, the test assist program causing the computer to carry out: an acquiring process of acquiring utterance information which indicates content of an utterance of a patient during a test performed on the patient, state information regarding feelings of the patient during the test, and basic information regarding the test; an interruption predicting process of predicting a probability of interruption of the test performed on the patient, from the utterance information, the state information, and the basic information, with use of a prediction model for predicting a probability of interruption of a test, the prediction model being generated by machine learning in which samples of past test interruptions are used as training data; and an outputting process of outputting the probability of interruption.
